1. If [x] stands for the greatest integer function, then the value of $\left[\frac{1}{2}+\frac{1}{1000}\right]+\left[\frac{1}{2}+\frac{2}{1000}\right]+.....+\left[\frac{1}{2}+\frac{999}{1000}\right]$:

यदि [x]; x के लिए महत्तम पूर्णांक फलन को दर्शाता है, तब $\left[\frac{1}{2}+\frac{1}{1000}\right]+\left[\frac{1}{2}+\frac{2}{1000}\right]+.....+\left[\frac{1}{2}+\frac{999}{1000}\right]$:

(A) 498 (B) 499 (C) 500 (D) 501

Ans. C

Sol. Every bracket is in format of $\left[\frac{1}{2}+\frac{k}{1000}\right]$

if $k=500 \Rightarrow \left[\frac{1}{2}+\frac{500}{1000}\right]=\left[\frac{1}{2}+\frac{1}{2}\right]=1$

So, from 500 to 999 evry bracket become 1.

Therefore ans is 500 (due to 500 times one)

2. If $f(x,y)=\left(\max(x,y)\right)^{\min(x,y)}$ and $g(x,y)=\max(x,y)-\min(x,y)$, then $f\left(g\left(-1,-\frac{3}{2}\right), g(-4,-1.75)\right)$ equals:

यदि $f(x,y)=\left(\max(x,y)\right)^{\min(x,y)}$ तथा $g(x,y)=\max(x,y)-\min(x,y)$, तो $f\left(g\left(-1,-\frac{3}{2}\right), g(-4,-1.75)\right)$ का मान होगा :

(A) – 0.5 (B) 0.5 (C) 1 (D) 1.5

Ans. D

Sol. As $g(x,y)=\max(x,y)-\min(x,y)$

$g\left(-1,-\frac{3}{2}\right)=-1-\left(-\frac{3}{2}\right)=\frac{3}{2}-1=\frac{1}{2}$

$g(-4,-1.75)=-1.75-(-4)=4-1.75=2.25=\frac{9}{4}$

$$f\left(\frac{1}{2},\frac{9}{4}\right)=\left(\frac{9}{4}\right)^{\frac{1}{2}}=\frac{3}{2}=1.5$$

3. Domain of $y=\log_{10}\left(\frac{5x-x^2}{4}\right)$:

$y=\log_{10}\left(\frac{5x-x^2}{4}\right)$ का प्रान्त है :

(A) (0, 5) (B) [1, 4] (C) $(-\infty,0)\cup(5,\infty)$ (D) $(-\infty, 1)\cup(4,\infty)$

Ans. A

Sol. $\frac{5x-x^2}{4}>0\Rightarrow x^2-5x<0$

$x(x-5)<0$

$x\in(0,5)$

4. The domain of definition of $f(x)=\frac{\sqrt{-\log_{0.3}(x-1)}}{\sqrt{x^2+2x+8}}$ is :

$f(x)=\frac{\sqrt{-\log_{0.3}(x-1)}}{\sqrt{x^2+2x+8}}$ का प्रान्त है :

(A) (1, 4) (B) (–2, 4) (C) (2, 4) (D) [2, ∞)

Ans. D

Sol. For domain $-\log_{0.3}(x-1)\geq 0$ and $x^2+2x+8>0$

$\Rightarrow \log_{0.3}(x-1)\leq 0$ $\Rightarrow (x+1)^2+7>0$

$\Rightarrow (x-1)\geq 1$ $\Rightarrow x\in R$

$\Rightarrow x\geq 2$ $\therefore$ Taking intersection $x\in[2,\infty)$

Hindi प्रान्त के लिए $-\log_{0.3}(x-1)\geq 0$ तथा $x^2+2x+8>0$

$\Rightarrow \log_{0.3}(x-1)\leq 0$ $\Rightarrow (x+1)^2+7>0$

$\Rightarrow (x-1)\geq 1$ $\Rightarrow x\in R$

$\Rightarrow \quad x \geq 2 \qquad\qquad \therefore \quad x \in [2, \infty)$

5. The domain of the function $f(x) = {}^{24-x}C_{3x-1} + {}^{40-6x}C_{8x-10}$ is :

(A) {2, 3} (B) {1, 2, 3} (C) {1, 2, 3, 4} (D) None of these

फलन $f(x) = {}^{24-x}C_{3x-1} + {}^{40-6x}C_{8x-10}$ का प्रान्त है :

(A) {2, 3} (B) {1, 2, 3} (C) {1, 2, 3, 4} (D) इनमें से कोई नहीं

Ans. A

Sol. Let $f(x) = g(x) + p(x)$

$g(x) = {}^{24-x}C_{3x-1}$

(i) $24 - x \geq 1$ (ii) $3x - 1 \geq 0$ (iii) $r \leq n$

$x - 23 \leq 0$ $x \geq \frac{1}{3}$ $3x - 1 \leq 24 - x$

$x \leq 23$ $x \in \left[\frac{1}{3}, \infty\right)$ $4x \leq 25$

So $x \in (-\infty, 23]$ $4x \leq 25$

$x \leq \frac{25}{4} \Rightarrow x \in \left(-\infty, \frac{25}{4}\right]$

from intersection of (i), (ii) and (iii)

$x \in \left[\frac{1}{3}, \frac{25}{4}\right]$ As $24 - x \in N$ and $3x - 1$ can be W.

So only integer value of x should accept.

Therefore $D_g \in \{1, 2, 3, 4, 5, 6\}$

$P(x) = {}^{40-6x}C_{8x-10}$

(i) $40 - 6x \geq 1$ (ii) $8x - 10 \geq 0$ (iii) $40 - 6x \geq 8x - 10$

$39 - 6x \geq 0$ $4x - 5 \geq 0$ $50 \geq 14x$

$13 - 2x \geq 0$ $x \geq \frac{5}{4}$ $x \leq \frac{50}{14} \sim 3.....$

$2x - 13 \leq 0$

$x \in \left(-\infty, \frac{13}{2}\right]$ $x \in \left[\frac{5}{4}, \infty\right)$ $x \in \left(-\infty, \frac{50}{14}\right]$

Intersection of (i), (ii) and (iii)

$x \in \left[\frac{5}{4}, \frac{25}{7}\right]$

Therefore $D_p = \{2,3\}$

$D_f = D_g \cap D_p = \{1,2,3,4,5,6\} \cap \{2,3\} = \{2,3\}$

6. The set of points for which $f(x) = \cos(\sin x) > 0$ contains :

(A) $(-\infty, 0]$ (B) $[-1, 1]$ (C) $(-\infty, \infty)$ (D) All are correct

उन बिन्दुओं का समुच्चय जिनके लिये $f(x) = \cos(\sin x) > 0$ निम्न को समाहित करता है :

(A) $(-\infty, 0]$ (B) $[-1, 1]$ (C) $(-\infty, \infty)$ (D) सभी सही है

Ans. D

Sol. $\cos(\sin x) > 0$

$\cos[-1,1] > 0$ $1^c \sim 57^o16'22''$

$\cos[-57^o, 57^o] > 0$

always true for every value of $x \Rightarrow x \in R$

so all sets are subsets of R.

7. The function $f(x) = \sqrt{\log_{10} \cos(2\pi x)}$ exists :

(A) for any rational x (B) only when x is a positive integer

(C) only when x is fractional (D) for any integer value of x including zero

फलन $f(x) = \sqrt{\log_{10} \cos(2\pi x)}$ का अस्तित्व है :

(A) x के किसी भी परिमेय मान के लिए (B) केवल तब, जबकि x धनात्मक पूर्णांक हो

(C) केवल तब, जबकि x भिन्नांश हो (D) शून्य सहित x के किसी भी पूर्णांक मान के लिए

Ans. D

Sol. $\cos(2\pi x) > 0$ and $\log_{10}(\cos 2\pi x) \geq 0$

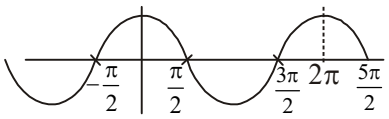
$\cos(2\pi x) \geq 1$

it can't be greater than but can be equal to 1 at every integer.

$x \in I$

$$x \in \left[0, \frac{\pi}{2}\right] \cup \left[\frac{3\pi}{2}, 2\pi\right]$$

$$x \in \left[2n\pi, 2n\pi + \frac{\pi}{2}\right] \cup \left[2n\pi + \frac{3\pi}{2}, 2n\pi + 2\pi\right]$$



Intersection of both is integers along with zero.

$$f(x) = \sqrt{\log(\cos(\sin x))}$$

$\log(\cos(\sin x)) \geq 0$

$\cos(\sin x) \geq 1$

but $\cos(\sin x) \leq 1$

$\sin x = 0$

$x = 0$

Therefore, domain : $x \in 0$

thus, Range : $R \in f(0) = 1$

8. The domain of the definition of $f(x) = \log\{(\log x)^2 - 5\log x + 6\}$ is equal to :

फलन $f(x) = \log\{(\log x)^2 - 5\log x + 6\}$ का प्रान्त है :

(A) $(0, 10^2)$ (B) $(10^3, \infty)$ (C) $(10^2, 10^3)$ (D) $(0, 10^2) \cup (10^3, \infty)$

Ans. D

Sol. $\log_{10}^2 x - 5\log_{10} x + 6 > 0$ Let $\log_{10} x = t \Rightarrow x = 10^t$

$$t^2 - 5t + 6 > 0 \Rightarrow (t-3)(t-2) > 0 \Rightarrow t \in (-\infty, 2) \cup (3, \infty)$$

$$x = 10^t \Rightarrow x \in \left(10^{-\infty}, 10^2\right) \cup \left(10^3, 10^{\infty}\right)$$

$$x \in \left(0, 10^2\right) \cup \left(10^3, \infty\right)$$

9. Domain of $y = \sqrt{-\log_{\frac{x+4}{2}}\left(\log_2 \frac{2x-1}{3+x}\right)}$:

$y = \sqrt{-\log_{\frac{x+4}{2}}\left(\log_2 \frac{2x-1}{3+x}\right)}$ का प्रान्त है :

(A) $(-4,-3) \cup (4, \infty)$ (B) $(-\infty, -3) \cup (4, \infty)$ (C) $(-\infty, -4) \cup (3, \infty)$ (D) $(-4,-3) \cup (3, 4)$

Ans. A

Sol. $f(x) = \sqrt{-\log_{\frac{x+4}{2}}\left(\log_2 \frac{2x-1}{3+x}\right)}$. For domain : $\log_{\frac{x+4}{2}} \frac{x+4}{2}\left(\log_2 \frac{2x-1}{3+x}\right) \le 0$

Case-I $0 < \frac{x+4}{2} < 1 \Rightarrow \quad -4 < x < -2$ .....A

then $\log_{\frac{x+4}{2}}\left(\log_2 \frac{2x-1}{3+x}\right) \le 0 \quad \Rightarrow \quad \log_2 \frac{2x-1}{3+x} \ge 1$

$\Rightarrow \quad \frac{2x-1}{3+x} \ge 2 \quad \Rightarrow \quad x < -3$ .....B

$\Rightarrow \quad$ on $A \cap B \quad x \in (-4,-3)$ .....(i)

Case-II $\frac{x+4}{2} > 1$ or $x > -2$ .....A

$\log_{\frac{x+4}{2}}\left(\log_2 \frac{2x-1}{3+x}\right) \le 0 \quad \Rightarrow \quad \log_2 \frac{2x-1}{3+x} \ge 1 \quad \Rightarrow \quad 1 < \frac{2x-1}{3+x} \le 2$

$\Rightarrow \quad x \in (4, \infty)$ .....(ii)

$\therefore \quad$ (i) $\cup$ (ii) Domain $x \in (-4,-3) \cup (4, \infty)$

Hindi. $f(x) = \sqrt{-\log_{\frac{x+4}{2}}\left(\log_2 \frac{2x-1}{3+x}\right)}$. प्रान्त के लिए : $\log_{\frac{x+4}{2}} \frac{x+4}{2}\left(\log_2 \frac{2x-1}{3+x}\right) \le 0$

स्थिति-I $0 < \frac{x+4}{2} < 1 \Rightarrow \quad -4 < x < -2$ .....A

तब $\log_{\frac{x+4}{2}}\left(\log_2 \frac{2x-1}{3+x}\right) \le 0 \Rightarrow \quad \log_2 \frac{2x-1}{3+x} \ge 1$

$\Rightarrow \quad \frac{2x-1}{3+x} \ge 2 \quad \Rightarrow \quad x < -3$ .....B

$\Rightarrow \quad A \cap B \quad x \in (-4,-3)$ .....(i)

स्थिति-II $\frac{x+4}{2} > 1$ या $x > -2$ .....A

$$\log_{\frac{x+4}{2}}\left(\log_2\frac{2x-1}{3+x}\right)\le 0 \quad\Rightarrow\quad \log_2\frac{2x-1}{3+x}\ge 1 \quad\Rightarrow\quad 1<\frac{2x-1}{3+x}\le 2$$

$\Rightarrow$ $x\in(4,\infty)$ .....(ii)

$\therefore$ (i) $\cup$ (ii) प्रांत $x\in(-4,-3)\cup(4,\infty)$

10. The domain of definition of $f(x)=\frac{\log_2(x+3)}{x^2+3x+2}$ is :

$f(x)=\frac{\log_2(x+3)}{x^2+3x+2}$ का प्रान्त है :

(A) $\frac{R}{\{-1,-2\}}$ (B) $(-2,\infty)$ (C) $\frac{R}{\{-1,-2,-3\}}$ (D) $\frac{(-3,\infty)}{\{-1,-2\}}$

Ans. D

Sol. $x^2+3x+2\neq 0\Rightarrow(x+2)(x+1)\neq 0$

$\Rightarrow x\neq -1,-2$

$x+3>0$

$\Rightarrow x\in(-3,\infty)$ $\qquad D_f\in\frac{(-3,\infty)}{\{-1,-2\}}$

11. The value of the function $f(x)=\frac{x^2-3x+2}{x^2+x-6}$ lies in the interval :

(A) $(-\infty,\infty)-\left\{\frac{1}{5},1\right\}$ (B) $(-\infty,\infty)$ (C) $(-\infty,\infty)-\{1\}$ (D) None of these

फलन $f(x)=\frac{x^2-3x+2}{x^2+x-6}$ का मान निम्न अन्तराल में स्थित है :

(A) $(-\infty,\infty)-\left\{\frac{1}{5},1\right\}$ (B) $(-\infty,\infty)$ (C) $(-\infty,\infty)-\{1\}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $f(x)=\frac{(x-2)(x-1)}{(x+3)(x-2)}$

Here $x-2$ is a common factor.

$$R_f \in R - \left\{\frac{1}{1}, \frac{2-1}{2+3}\right\}$$

$$R_f \in R - \left\{\frac{1}{5}, 1\right\}$$

12. Let $f_n(\theta) = \dfrac{\cos\frac{\theta}{2} + \cos 2\theta + \cos\frac{7\theta}{2} + ... + \cos(3n-2)\frac{\theta}{2}}{\sin\frac{\theta}{2} + \sin 2\theta + \sin\frac{7\theta}{2} + ... + \sin(3n-2)\frac{\theta}{2}}$. Then match the entries of column-I with their corresponding values given in column-II.

माना $f_n(\theta) = \dfrac{\cos\frac{\theta}{2} + \cos 2\theta + \cos\frac{7\theta}{2} + ... + \cos(3n-2)\frac{\theta}{2}}{\sin\frac{\theta}{2} + \sin 2\theta + \sin\frac{7\theta}{2} + ... + \sin(3n-2)\frac{\theta}{2}}$ हो, तो कॉलम-I को कॉलम-II के साथ सुमेलित कीजिए :

Column–I	**Column-II**
(A) $f_3\left(\frac{3\pi}{16}\right)$	(P) $2-\sqrt{3}$
(B) $f_5\left(\frac{\pi}{28}\right)$	(Q) $\sqrt{3+2\sqrt{2}}$
(C) $f_7\left(\frac{\pi}{60}\right)$	(R) $\sqrt{2}-1$
	(S) $\sqrt{7+4\sqrt{3}}$

Ans. (A) R ; (B) Q ; (C) S

Sol.
$$f_n(\theta) = \frac{\sin\left((3n-2)\times\frac{3\theta}{2}\right)\times\left(\frac{(3n-2)\frac{\theta}{2}+\frac{\theta}{2}}{2}\right)}{\sin\left(\frac{\frac{3\theta}{2}}{2}\right)\times\sin\left((3n-2)\times\frac{3\theta}{2}\right)\times\sin\left(\frac{(3n-2)\frac{\theta}{2}+\frac{\theta}{2}}{2}\right)}{\sin\left(\frac{3\theta}{4}\right)}$$

$$f_n(\theta) = \cot\frac{(3n-1)\theta}{4}$$

(A) $f_3\left(\frac{3\pi}{16}\right) = \cot\frac{(3\times 3-1)}{4}\times\frac{3\pi}{16} = \cot\frac{3\pi}{8} = \cot 67\frac{1}{2}^{\circ} = \sqrt{2}-1$

(B) $f_5\left(\frac{\pi}{28}\right) = \cot\frac{14}{4}\times\frac{\pi}{28} = \cot\frac{\pi}{8} = \sqrt{3+2\sqrt{2}}$

(C) $f_7\left(\frac{\pi}{60}\right) = \cot\frac{20}{4}\times\frac{\pi}{60} = \cot\frac{\pi}{12} = \sqrt{7+4\sqrt{3}}$

1. Find the range of the following function: $f(x)=\log_2\left(\frac{\sin x-\cos x+3\sqrt{2}}{\sqrt{2}}\right)$.

निम्न फलन का परिसर ज्ञात कीजिये : $f(x)=\log_2\left(\frac{\sin x-\cos x+3\sqrt{2}}{\sqrt{2}}\right)$

Ans. $[1,2]$

Sol. $f(x)=\log_2\left[\frac{\left[-\sqrt{2},\sqrt{2}\right]+3\sqrt{2}}{\sqrt{2}}\right]$

$f(x)=\log_2\left[\frac{2\sqrt{2}}{\sqrt{2}},\frac{4\sqrt{2}}{\sqrt{2}}\right]$

$f(x)=\log_2[2,4]$

$R_f\in[1,2]$

2. Find the domain and range of the function f(x) = $\sqrt{2-x}+\sqrt{1+x}$

फलन f(x) = $\sqrt{2-x}+\sqrt{1+x}$ के प्रान्त तथा परिसर ज्ञात कीजिये।

Ans. $[\sqrt{3}\,,\sqrt{6}\,]$

Sol. for domain

$2-x\ge 0$ and $1+x\ge 0$

$x-2\le 0$ and $x\ge -1$

for range

$f'(x)=\frac{1}{2\sqrt{2-x}}+\frac{1}{2\sqrt{1+x}}=0$

$f'(x)=0\Rightarrow\frac{1}{\sqrt{2-x}}=\frac{-1}{\sqrt{1+x}}$

$x \in (-\infty, 2]$ and $x \in [-1, \infty)$ squaring both sides

$$\frac{1}{2-x} = \frac{1}{1+x} \Rightarrow 1+x = 2-x$$

$$2x = 1 \Rightarrow x = \frac{1}{2}(\text{maxima})$$

$$R_f \in \left[f(-1) \text{ or } f(2); f\left(\frac{1}{2}\right) \right]$$

$$= \left[\sqrt{3}, 2\sqrt{\frac{3}{2}} \right] = \left[\sqrt{3}, \sqrt{\frac{4\times 3}{2}} \right]$$

So, $R_f \in \left[\sqrt{3}, \sqrt{6} \right]$

3. Find the range of the following function $f(x) = \frac{3}{2-x^2}$.

फलन $f(x) = \frac{3}{2-x^2}$ का परिसर ज्ञात कीजिये।

Ans. $(-\infty, 0) \cup [3/2, \infty)$

Sol. $f(x) = x^3 + x^2 + 3x + \sin x, x \in R \quad \Rightarrow \quad f'(x) = 3x^2 + 2x + 3 + \cos x$

$\because \quad 3x^2 + 2x + 3 \geq \frac{32}{12}$ as $a = 3 > 0$ and $D < 0 \quad \Rightarrow \quad -1 \leq \cos x \leq 1$

so $f'(x) > 0 \ \forall \ x \in R \quad \Rightarrow \quad \lim_{x\to\infty} f(x) = +\infty \Rightarrow \quad \lim_{x\to-\infty} f(x) = -\infty$

Hence f(x) is one-one and onto function (as f(x) is continuous function)

Hindi. $f(x) = x^3 + x^2 + 3x + \sin x, x \in R \quad \Rightarrow \quad f'(x) = 3x^2 + 2x + 3 + \cos x$

$\because \quad 3x^2 + 2x + 3 \geq \frac{32}{12}$ ($\because a = 3 > 0$ तथा $D < 0$) $\Rightarrow \quad -1 \leq \cos x \leq 1$

अतः $f'(x) > 0 \ \forall \ x \in R \quad \Rightarrow \quad \lim_{x\to\infty} f(x) = +\infty \Rightarrow \quad \lim_{x\to-\infty} f(x) = -\infty$

अतः f(x) एकैकी–आच्छादक फलन है ($\because$ f(x) सतत् फलन है)

4. The range of the function $y = \frac{1}{2-\sin 3x}$ is :

(A) $\left(\frac{1}{3}, 1\right)$ (B) $\left[\frac{1}{3}, 1\right)$ (C) $\left[\frac{1}{3}, 1\right]$ (D) None of these

फलन $y = \frac{1}{2 - \sin 3x}$ का परिसर है :

(A) $\left(\frac{1}{3}, 1\right)$ (B) $\left[\frac{1}{3}, 1\right)$ (C) $\left[\frac{1}{3}, 1\right]$ (D) इनमें से कोई नहीं

Ans. C

Sol. $y = \frac{1}{2 - [-1,1]}$

$y = \frac{1}{[2+1, 2-1]} = \frac{1}{[1,3]}$

$y = \left[\frac{1}{3}, 1\right]$

5. Find the range of the following function, $y = \log_{\sqrt{7}}(\sqrt{2}(\sin x - \cos x) + 5)$:

फलन $y = \log_{\sqrt{7}}(\sqrt{2}(\sin x - \cos x) + 5)$ का परिसर है :

(A) R (B) Z (C) $[\log_7 4, \log_7 5]$ (D) $[2\log_7 3, 2]$

Ans. D

Sol. $= \log_{\sqrt{7}}\left(\sqrt{2}\left[-\sqrt{2}, \sqrt{2}\right] + 5\right)$

$= \log_{\sqrt{7}}\left([-2, 2] + 5\right)$

$= \log_{\sqrt{7}}(3, 7)$

$R_f \in [2\log_7 3, 2]$

6. Which of the following function (s) has the range [-1, 1] :

निम्न में से किस फलन का परिसर [-1, 1] है :

(A) $f(x) = \cos(2\sin x)$ (B) $g(x) = \cos\left(1 - \frac{1}{1+x^2}\right)$

(C) $h(x) = \sin(\log_2 x)$ (D) $k(x) = \tan(e^x)$

Ans. C

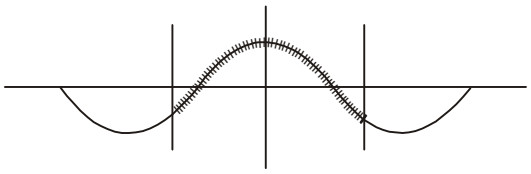
Sol. (A) $f(x) = \cos[-2,2]$

$R_f \in \left[\cos(-2), 1\right]$ or $\left[\cos 2^c, 1\right]$

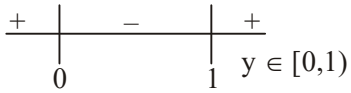
(B) $g(x) = \cos\left(1 - \frac{1}{1+x^2}\right) = \cos\left(\frac{1+x^2-1}{1+x^2}\right) = \cos\left(\frac{x^2}{1+x^2}\right)$

first find range of $\frac{x^2}{1+x^2}$. So, $y + yx^2 = x^2 \Rightarrow y = x^2(1-y)$

$x = \sqrt{\frac{y}{1-y}}$ $\frac{y}{1-y} \geq 0 \Rightarrow \frac{y}{y-1}$



So, $1 - y \neq 0$, $y \neq 1$



$y \in [0,1)$

Now, $g(x) = \cos[0,1)$

$R_g \in \left[\cos 0, \cos 1\right) \Rightarrow R_g \in \left(\frac{1}{2}, 1\right]$

(C) $h(x) = \sin(\log_2 x) = \sin(0, \infty) = [-1,1]$

(D) $k(x) = \tan(e^x) = \tan(0, \infty) = (-\infty, \infty)$

7. If $A = \left\{x : \frac{\pi}{6} \leq x \leq \frac{\pi}{3}\right\}$ and $f(x) = \cos x - x(1+x)$ then $f(A)$ is equal to :

यदि $A = \left\{x : \frac{\pi}{6} \leq x \leq \frac{\pi}{3}\right\}$ तथा $f(x) = \cos x - x(1+x)$ तब $f(A)$ बराबर है :

(A) $\left[\frac{\pi}{6}, \frac{\pi}{3}\right]$ (B) $\left[-\frac{\pi}{3}, -\frac{\pi}{6}\right]$

(C) $\left[\frac{1}{2} - \frac{\pi}{3}\left(1+\frac{\pi}{3}\right), \frac{\sqrt{3}}{2} - \frac{\pi}{6}\left(1+\frac{\pi}{6}\right)\right]$ (D) $\left[\frac{1}{2} + \frac{\pi}{3}\left(1-\frac{\pi}{3}\right), \frac{\sqrt{3}}{2} + \frac{\pi}{6}\left(1-\frac{\pi}{6}\right)\right]$

Ans. C

Sol. Given : $f(x) = \cos x - x(1+x)$ ...(i)

if $\frac{\pi}{6} \le x \le \frac{\pi}{3}$ or $x \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right]$ direct substitute this in function

So, $R_f\left[\cos\frac{\pi}{3} - \frac{\pi}{3}\left(1+\frac{\pi}{3}\right), \cos\frac{\pi}{6} - \frac{\pi}{6}\left(1+\frac{\pi}{6}\right)\right]$

8. If $f(x) = [x^2] - [x]^2$ where $[\cdot]$ denotes the greatest integer function and $x \in [0, 2]$, the set of values of $f(x)$ is :

यदि $f(x) = [x^2] - [x]^2$ जहाँ $[\cdot]$ महत्तम पूर्णांक फलन को निरूपित करता है तथा $x \in [0, 2]$, तब $f(x)$ के मानों का समुच्चय है :

(A) $\{-1, 0\}$ (B) $\{-1, 0, 1\}$ (C) $\{0\}$ (D) $\{0, 1, 2\}$

Ans. D

Sol.
$$g(x) = \left[x^2\right] = \begin{cases} 0 & 0 \le x < 1 \Rightarrow 0 \le x^2 < 1 \\ 1 & 1 \le x < \sqrt{2} \Rightarrow 1 \le x^2 < 2 \\ 2 & \sqrt{2} \le x\sqrt{3} \Rightarrow 2 \le x^2 < 3 \\ 3 & \sqrt{3} \le x < 2 \Rightarrow 3 \le x^2 < 4 \\ 4 & x = 2 \Rightarrow x^2 = 4 \end{cases}$$

$$p(x) = \left[x^2\right] = \begin{cases} 0 & 0 \le x < 1 \\ 1 & 1 \le x < 2 \\ 4 & x = 2 \end{cases}$$

Now, $f(x) = \begin{cases} 0; & 0 \le x < 1 \,\&\, x = 2 \,\&\, x = 1 \\ 1; & \sqrt{2} \le x < \sqrt{3} \\ 2; & \sqrt{3} \le x < 2 \end{cases}$

So, $R_f \in \{0,1,2\}$

9. Range of the function $f(x) = \frac{\{x\}}{1+\{x\}}$ where $\{x\}$ denotes the fractional part function is :

फलन $f(x) = \frac{\{x\}}{1+\{x\}}$ का परिसर जहां $\{x\}$ भिन्नात्मक फलन है :

(A) [0 , 1) (B) $\left[0,\frac{1}{2}\right]$ (C) $\left[0,\frac{1}{2}\right)$ (D) $\left(0,\frac{1}{2}\right)$

Ans. C

Sol. $y=\frac{\{x\}}{1+\{x\}}$

$y+\{x\}\times y=\{x\}$

$y=\{x\}\,(1-y)$

$\{x\}=\frac{y}{1-y}$

$0\le\frac{y}{1-y}\le 1$

for I_1 $\frac{y}{1-y}\ge 0$

$\frac{y}{y-1}\le 0$

$y\in[0,1)$

for I_2 $\frac{y}{1-y}-1\le 0$

$\frac{y-(1-y)}{1-y}\le 0$

$\frac{y-1+y}{1-y}\le 0$

$\frac{2y-1}{y-1}\ge 0$

$y\in\left(-\infty,\frac{1}{2}\right]\cup(1,\infty)$

intersection of I_1 & I_2

$R_f\in\left[0,\frac{1}{2}\right]$

10. For the function $f(x)=\frac{e^x+1}{e^x-1}$, if $n(d)$ denotes the number of integers which are not in its domain and $n(r)$ denotes the number of integers which are not in its range, then $n(d)+n(r)$ is equal to :

(A) 2 (B) 3 (C) 4 (D) Infinite

फलन $f(x) = \dfrac{e^x + 1}{e^x - 1}$, के लिये यदि $n(d)$ पूर्णाकों की संख्या निरूपित करता है जो प्रान्त में नहीं है तथा $n(r)$ पूर्णाकों की संख्या निरूपित करता है जो परिसर में नहीं है तो $n(d) + n(r)$ का मान होगा :

(A) 2 (B) 3 (C) 4 (D) अनन्त

Ans. C

Sol. $e^x - 1 \neq 0$ $e^x \neq 1$ $x \neq 0$

$D_f \in R - \{0\}$ $n(d) = 0$

$$y = \frac{e^x + 1}{e^x - 1} \Rightarrow \frac{y+1}{y-1} = \frac{e^x + 1 + e^x - 1}{e^x + 1 - e^x + 1}$$

$$\frac{y+1}{y-1} = e^x \Rightarrow x = \ell n\left(\frac{y+1}{y-1}\right)$$

for range $\dfrac{y+1}{y-1} > 0$

$x \in (-\infty, -1) \cup (1, \infty)$ or $R - [-1,1]$

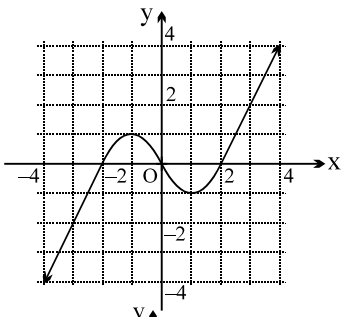
$n(r) = -1,0,1$

Total no. of integer = 4

11. The graph of the function $y = g(x)$ is shown.

The number of solutions of the equation $\left|\,|g(x)| - 1\,\right| = \dfrac{1}{2}$, is

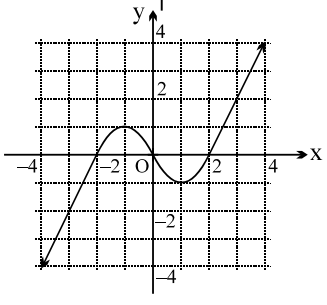
(A) 4 (B) 5
(C) 6 (D) 8



$y = g(x)$ का आरेख चित्रानुसार दिखाया गया है तो समीकरण

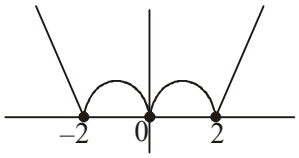
$\left|\,|g(x)| - 1\,\right| = \dfrac{1}{2}$ के कुल हलों की संख्या होगी–

(A) 4 (B) 5
(C) 6 (D) 8

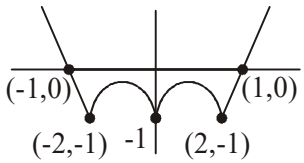


Ans. D

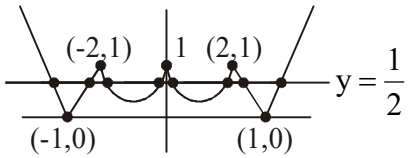
Sol. $|g(x)| \Rightarrow$



$|g(x)|-1 \Rightarrow$



$||y(x)|-1| =$



Total number of solution = 8

12. Column-I contains rational algebraic expressions and column-II contains possible integers which lie in their range. Match the entries of column-I with one or more entries of the elements of column-II :

स्तम्भ -I में बीजिय व्यजंक दिये गये है तथा स्तम्भ -II में पूर्णांक दिये गये है जो व्यंजको के परिसर में विद्यमान है। स्तम्भ -I के प्रविष्टियों को स्तम्भ -II की प्रविष्टियों से मिलान करवायें :

Column-I	**Column-II**
(A) $y = \frac{x^2-2x+4}{x^2+2x+4},\ x \in R$	(P) 1
(B) $y = \frac{x^2-3x-2}{2x-3},\ x \in R$	(Q) 4
(C) $y = \frac{2x^2-2x+4}{x^2-4x+3},\ x \in R$	(R) – 3
	(S) – 10

Ans. (A) P,(B) P,Q,R,S (C) P,Q,S

Sol. (A) $y = \frac{x^2-2x+4}{x^2+2x+4}$

$x^2y + 2xy + 4y - x^2 + 2x - 4 = 0$

$x^2 (y - 1) + x(2y + 2) + (4y - 4) = 0$

C–I : L.C. $\neq 0$, $D \geq 0$

$y \neq 1$, $b^2 - 4ac \geq 0$

$4 (y + 1)^2 - 4(y - 1) \times 4 (y - 1) \geq 0$

$(y + 1)^2 - 4(y - 1)^2 \geq 0$

$y^2 + 2y + 1 - 4y^2 + 8y - 4 \geq 0$

$-3y^2 + 10y - 3 \geq 0$

$3y^2 - 10y + 3 \leq 0$

$3y^2 - 9y - y + 3 \leq 0$

$(3y - 1) (y - 3) \leq 0$

$$\Rightarrow y \in \left[\frac{1}{3}, 3\right] - \{1\}$$

C–II : L.C. $= 0 \Rightarrow y = 1$

Put this in question

$$1 = \frac{x^2 - 2x + 4}{x^2 + 2x + 4}$$

$x^2 + 2x + 4 = x^2 - 2x + 4$

$4x = 0$

$x = 0$

So $y = 1$ will be in range

$$R_f \in \left[\frac{1}{3}, 3\right] \text{ [only (P) lie in range]}$$

(B) $2xy - 3y = x^2 - 3x - 2$

$x^2 - x(2y + 3) + 3y - 2 = 0$

C–I : L.C. $\neq 0$, $\Delta \geq 0$

$1 \neq 0$, $b^2 - 4ac \geq 0$

(only this case will be possible)

L.C. $= 0$ not possible

$(2y + 3)^2 - 4(3y - 2) \geq 0$

$4y^2 + 9 + 12y - 12y + 8 \geq 0$

$4y^2 + 17 \geq 0$

it is always true for every real value of y.

$R_f \in R$

P, Q, R and S all lie in range.

(C) $y = \dfrac{2x^2 - 2x + 4}{x^2 - 4x + 3}$

$x^2y - 4xy + 3y - 2x^2 + 2x - 4 = 0$

$x^2(y - 2) - x(4y - 2) + (3y - 4) = 0$

C–I : L.C. $\neq 0$, $\Delta \neq 0$

$y \neq 2$, $b^2 - 4ac \geq 0$

$4(2y - 1)^2 - 4(y - 2)(3y - 4) \geq 0$

$4y^2 - 4y + 1 - [3y^2 - 4y - 6y + 8] \geq 0$

$y^2 + 6y - 7 \geq 0$

$y^2 + 7y - y - 7 \geq 0$

$(y + 7)(y - 1) \geq 0$

$\Rightarrow y \in (-\infty, -7] \cup [1, \infty) - \{2\}$

C–II : if L.C. $= 0 \Rightarrow y = 2$

$$2 = \frac{2x^2 - 2x + 4}{x^2 - 4x + 3}$$

$2x^2 - 8x + 6 = 2x^2 - 2x + 4$

$2 = 6x$

$$x = \frac{1}{3}$$

So, $y = 2$ will be in range

$\therefore R_f \in (-\infty, -7] \cup [1, \infty)$

P, Q and S lie in range.

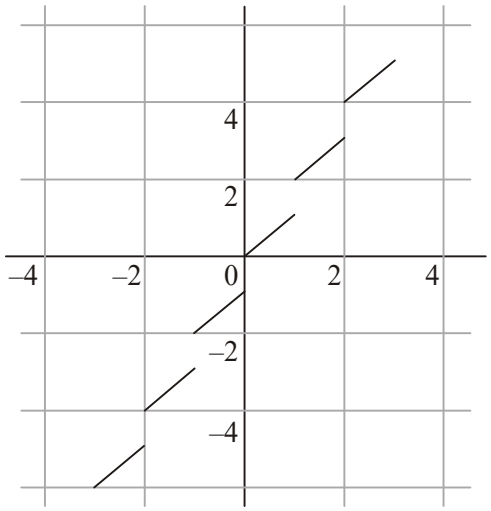
Que. (13 to 16) Draw graphs of the following function, where [ ] denotes the greatest integer function.

Que. (13 to 16) निम्न फलनो का ग्राफ खीचिए जहाँ [ ] महत्तम पूर्णांक फलन है।

13. $f(x) = x + [x]$

Ans.

Sol.



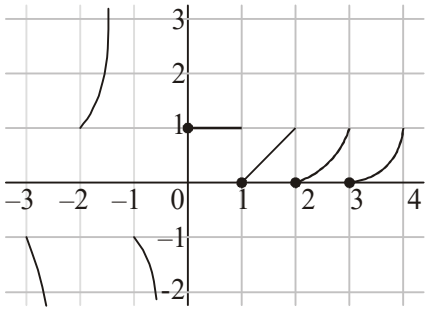
14. $y = (x)^{[x]}$ where $x = [x] + (x)$ & $x > 0$ & $x \le 3$

Ans.

Sol. Given : $(x) = x - [x]$

$(x) = \{x\}$

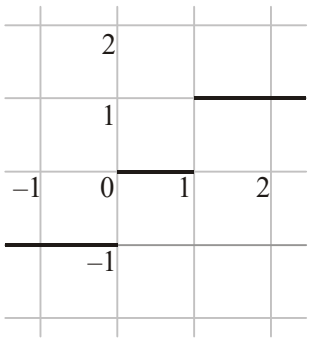
So, $y = \{x\}^{[x]}$



15. $y = \text{sgn}[x]$

Ans.

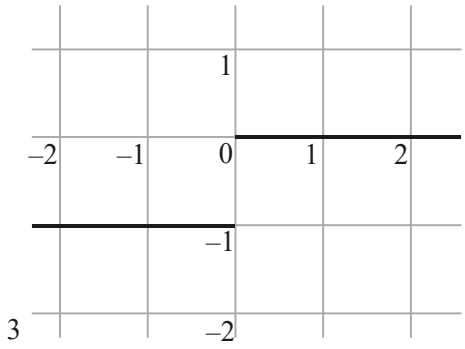
Sol.



16. $\operatorname{sgn}(x-|x|)$

Ans.

Sol.



1. Let $f: X \to Y$ be a function such that $f(x) = \sqrt{x-2} + \sqrt{4-x}$, then the set of X and Y for which f(x) is both injective as well as surjective, is :

(A) [2, 4] and $[\sqrt{2}, 2]$ (B) [3, 4] and $[\sqrt{2}, 2]$ (C) [2, 4] and [1, 2] (D) [2, 3] and [1, 2]

माना $f: X \to Y$ में परिभाषित फलन $f(x) = \sqrt{x-2} + \sqrt{4-x}$, एकैकी आच्छादक फलन हो तो X तथा Y के मानों का समुच्चय होगा :

(A) [2, 4] तथा $[\sqrt{2}, 2]$ (B) [3, 4] तथा $[\sqrt{2}, 2]$ (C) [2, 4] तथा [1, 2] (D) [2, 3] तथा [1, 2]

Ans. B

Sol. $f(x) = \sqrt{x-2} + \sqrt{4-x}$

For the square root to exist,

$x - 2 \geq 0 \Rightarrow x \geq 2$

$4 - x \geq 0 \Rightarrow x \leq 4$

So, $D_f \in [2,4]$

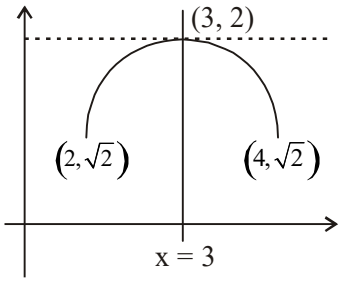
Now,

$f'(x) = 0$

$\Rightarrow \frac{1}{2\sqrt{x-2}} - \frac{1}{2\sqrt{4-x}} = 0$

$\Rightarrow (x-2) = 4 - x$

$\Rightarrow x = 3$



$f(3) = 2,\ f(2) = \sqrt{2},\ f(4) = \sqrt{2}$

The graph of the function,

$\therefore \text{Range}\left[\sqrt{2}, 2\right]$

Now, for the function to be both injective and surjective the domain has to be restricted in either [2,3] or [3,4].

Hence, from the given options,

$X = [3,4],\ Y = \left[\sqrt{2}, 2\right]$

2. Which of the following function is surjective but not injective :

निम्न फलनों में से कौन से आच्छादक है परन्तु एकैकी नही है :

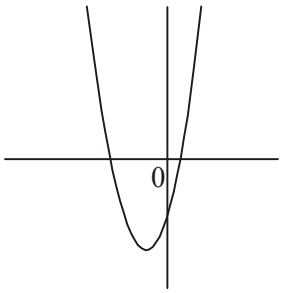
(A) $f : R \to R\ f(x) = x^4 + 2x^3 - x^2 + 1$ (B) $f : R \to R\ f(x) = x^3 + x + 1$

(C) $f : R \to R^+\ f(x) = \sqrt{1 + x^2}$ (D) $f : R \to R\ f(x) = x^3 + 2x^2 - x + 1$

Ans. D

Sol. (A) $f(x) = x^4 + 2x^3 - x^2 + 1$

is not an onto function (even degree polynomial)



(B) $f(x) = x^3 + x + 1$

is one-one and onto both.

(c) $f(x) = \sqrt{1 + x^2}$

$R_f \in [1, \infty)$ but codomain is R^+. So it is not onto function. Although its one-one.

(D) $f(x) = x^3 + 2x^2 - x + 1$

$f'(x) = 3x^2 + 4x - 1$

draw the graph of $3x^2 + 4x - 1$

clearly in graph $\frac{dy}{dx} < 0$ and $\frac{dy}{dx} > 0$.

So the f(x) is not one-one function.

3. Let $f : R \to R$ be a function defined by $f(x) = \frac{x^2 + 2x + 5}{x^2 + x + 1}$ is :

(A) one-one and into (B) one-one and onto

(C) many-one and onto (D) many-one and into

माना $f : R \to R$ परिभाषित है $f(x) = \frac{x^2 + 2x + 5}{x^2 + x + 1}$ तो फलन :

(A) एकैकी तथा अन्तःक्षेपी होगा (B) एकैकी तथा आच्छादक होगा

(C) बहुएकैकी तथा आच्छादक (D) बहुएकैकी तथा अन्तःक्षेपी

Ans. D

Sol. $\frac{\text{Quadratic}}{\text{Quadratic}}$ (with no common factor) is example of many-one.

Also range of given function is subset of real no.

Therefore the function is manyone-into.

4. The function $f : [2, \infty) \to Y$ defined by $f(x) = x^2 - 4x + 5$ is both one-one and onto if :

फलन $f : [2, \infty) \to Y$ परिभाषित है $f(x) = x^2 - 4x + 5$ एक एकैकी आच्छादक फलन है तो :

(A) $Y = R$ (B) $Y = [1, \infty)$ (C) $Y = [4, \infty)$ (D) $Y = [5, \infty)$

Ans. B

Sol. $f(x) = (x^2 - 4x + 4) + 1$

$f(x) = (x - 2)^2 + 1$

$R_f \in [1, \infty)$

for onto function

range = co-domain

$Y = R_f = [1, \infty)$

5. Let $f : R \to R$ be a function defined by $f(x) = x^3 + x^2 + 3x + \sin x$. Then f is :

(A) one- one and onto (B) one -one and into

(C) many one and onto (D) many one and into

माना फलन $f : R \to R$ इस प्रकार कि $f(x) = x^3 + x^2 + 3x + \sin x$ तो f होगा :

(A) एकैकी तथा आच्छादक होगा (B) एकैकी तथा अन्तःक्षेपी होगा

(C) बहुएकैकी तथा आच्छादक (D) बहुएकैकी तथा अन्तःक्षेपी

Ans. A

Sol. $f(x) = x^3 + x^2 + 3x + \sin x$

$\Rightarrow f'(x) = 3x^2 + 2x + 3 + \cos x \geq 3x^2 + 2x + 3 - 1$, since $-1 \leq \cos x \leq 1$

$\Rightarrow f'(x) \geq 3x^2 + 2x + 2$

Now discriminant of above quadratic is $= 2^2 - 4 \cdot 3 \cdot 2 < 0$

Hence $f'(x) > 0$ for all $x \in R$

Therefore f is one-one function, also every odd degree polynomial has it's range R. Thus f is an onto function.

6. If A be the set of all triangles and B that of positive real numbers, then the mapping $f : A \to B$ given by $f(\Delta)$ = area of Δ, $(\Delta \in A)$ is :

(A) one-one into mapping (B) one-one onto mapping

(C) many-one into mapping (D) many-one onto mapping

यदि A सभी त्रिभुजों का समुच्चय है तथा B सभी धनात्मक वास्तविक संख्याओं का समुच्चय है तथा फलन $f : A \to B$ इस प्रकार की $f(\Delta) = \Delta$ का क्षेत्रफल, $(\Delta \in A)$ तो f होगा :

(A) एकैकी तथा अन्तःक्षेपी होगा (B) एकैकी तथा आच्छादक होगा

(C) बहुएकैकी तथा अन्तःक्षेपी (D) बहुएकैकी तथा आच्छादक

Ans. D

Sol. Since the area of any triangle is a positive real number, therefore f-image of any triangle.

$\Delta \in A$, i.e. $f(\Delta) \in B$, corresponding to any real numbers in B, there is a triangle in A. Thus, f is onto.

Again there can be many triangles having the same area. Therefore, the mapping f is many-one.

Hence, the mapping f is many-one onto.

7. Let $f : R \to R$ and $g : R \to R$ be two one-one onto functions such that they are mirror image of each other about the line $y = 0$, then $h(x) = f(x) + g(x)$ is :

(A) one-one and onto (B) one-one but not onto

(C) not one-one but onto (D) Neither one-one nor onto

माना $f : R \to R$ तथा $g : R \to R$ दो एकैकी तथा आच्छादक फलन है इस प्रकार कि उनके आरेख रेखा $y = 0$ के सापेक्ष एक दूसरे के प्रतिबिम्ब है तो $h(x) = f(x) + g(x)$ होगा :

(A) एकैकी परन्तु आच्छादक नहीं (B) आच्छादक परन्तु एकैकी नहीं

(C) आच्छादक तथा एकैकी (D) न तो एकैकी न ही आच्छादक

Ans. D

Sol. Given that :-

$f : R \to R$ and $g : R \to R$

One-one and onto function and mirror image of each other about line $y = a$.

$h(x) = f(x) + g(x)$

To find :- h(x) is which type of function.

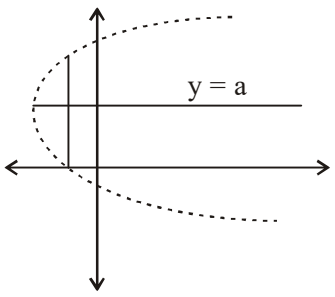
Solution :-

$f(x) = a + h$

$g(x) = a - h$

$f(x) + g(x) = h(x)$

$h(X) = a + h + a - h$



$h(x) = 2a$

$h(x)$ is constant.

So it is many one and into function.

It is neither one-one and nor onto.

8. Let $f : N \to N$ such that $f(n) = \begin{cases} \frac{n+1}{2}, & \text{if n is odd} \\ \frac{n}{2}, & \text{if n is even} \end{cases}$ for all $n \in N$ then f is :

(A) one-one but not onto
(B) onto but not one-one
(C) onto and one-one
(D) Neither one-one nor onto

माना $f : N \to N$ इस प्रकार की $f(n) = \begin{cases} \frac{n+1}{2}, & \text{य दि n विषम है} \\ \frac{n}{2}, & \text{य दि n सम है} \end{cases}$ सभी $n \in N$. तो f होगा :

(A) एकैकी परन्तु आच्छादक नहीं
(B) आच्छादक परन्तु एकैकी नहीं
(C) आच्छादक तथा एकैकी
(D) न तो एकैकी न ही आच्छादक

Ans. B

Sol. $f(n) = \left\{ \begin{array}{ll} \frac{n+1}{2} & \text{if n is odd} \\ \frac{n}{2} & \text{if n is even} \end{array} \right\}$ for all $n \in N$

$f : N \to N$ is defined as

It acan be observed that :

$f(1) = \frac{1+1}{2} = 1$ and $f(2) = \frac{2}{2} = 1$

$\therefore$ f(1) = f(2), where $1 \neq 2$

$\therefore$ f is not one-one.

Consider a natural number (n) in co-domain N case I : n is odd

$\therefore$ $n = 2r + 1$ for some $r \in N$.

Then, there exists $4r+1 \in N$ such that $f(4r+1)=\frac{4r+1+1}{2}=2r+1$

Case II : n is even

$\therefore$ n = 2r for some $r \in N$.

Then, there exists $4r \in N$ such that $f(4r)=\frac{4r}{2}=2r$

$\therefore$ f is onto.

Hence, f is not a bijective function.

9. Which of the following functions are not injective mapping-

(A) $f(x) = |x + 1|, x \in [-1, \infty)$ (B) $g(x) = x + \frac{1}{x}$; $x \in (0, \infty)$

(C) $h(x) = x^2 + 4x - 5$; $x \in (0, \infty)$ (D) $k(x) = e^{-x}$; $x \in [0, \infty)$

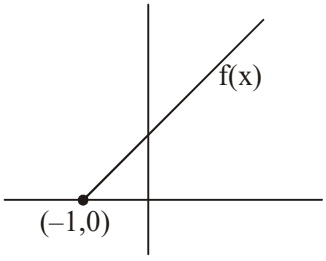
कौनसा फलन एकैकी नहीं है–

(A) $f(x) = |x + 1|, x \in [-1, \infty)$ (B) $g(x) = x + \frac{1}{x}$; $x \in (0, \infty)$

(C) $h(x) = x^2 + 4x - 5$; $x \in (0, \infty)$ (D) $k(x) = e^{-x}$; $x \in [0, \infty)$

Ans. B

Sol. (A) $f(x) = |x + 1|; x \in [-1, \infty)$



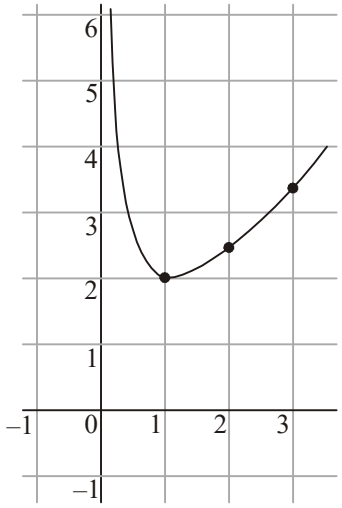
from grapoh of f(x);

Clearly f(x) is one-one.

(B) $f(x)=x+\frac{1}{x}; x \in (0,\infty)$

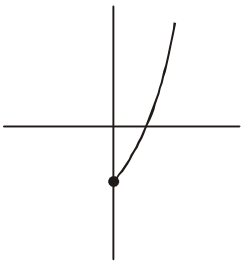
from graph of $x+\frac{1}{x}$ it is clear.

that f(x) is not injective.



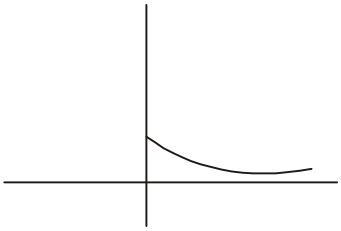
(C) $f(x) = x^2 + 4x - 5; x \in (0, \infty)$

from graph of f(x) it can be observe that f(x) is injective.



(D) $f(x) = e^{-x}; x \in [0, \infty)$

from graph f(x) is one-one function.



10. Let A = R – {2} and B = R – {1}. Let $f : A \to B$ be defined by $f(x) = \frac{x-3}{x-2}$ then-

(A) f is one-one (B) f is onto (C) f is bijective (D) None of these

माना A = R – {2} तथा B = R – {1}. फलन $f : A \to B$ इस प्रकार की $f(x) = \frac{x-3}{x-2}$ तो -

(A) f एकैकी होगा (B) f आच्छादक होगा (C) f व्युत्क्रमणीय होगा (D) कोई नहीं

Ans. ABC

Sol. Given :

$$f(x)=\frac{x-3}{x-2}$$

$D_f = R - \{2\}$

$R_f = R - \{1\} = B \Rightarrow$ Therefore f(x) is onto.

$$f'(x)=\frac{(x-2)1-(x-3)}{(x-2)^2}=\frac{x-2-x+3}{(x-2)^2}=\frac{1}{(x-2)^2}$$

$f'(x) > 0$ so f(x) is always increasing function.

Therefore f(x) is one-one function.

Hence f(x) is bijective function.

11. Let f: [–1, 1] $\rightarrow$ [0, 2] be a linear function which is onto then f(x) is/are.
(A) 1 – x (B) 1 + x (C) x –1 (D) x + 2
माना f: [–1, 1] $\rightarrow$ [0, 2] एक रेखीय फलन है जो आच्छादक है तो f(x) होगा –
(A) 1 – x (B) 1 + x (C) x –1 (D) x +2

Ans. AB

Sol. Option (A) $f(x) = 1 - x \ \forall \ x \in [-1,1]$

it is an always decreasing function.

$$\text{So, } \left.\begin{matrix}\max = 1-(-1)=2\\ \min = 1-(1)=0\end{matrix}\right] \Rightarrow R_f \in [0,2]$$

Therefore onto function.

Option (B) $f(x)=1+x \forall x \in [-1,1]$

f(x) is always increasing function.

$$\text{So } \left.\begin{matrix}\max = 1+(-1)=0\\ \min = 1+(1)=2\end{matrix}\right] \Rightarrow R_f \in [0,2]$$

Therefore onto function.

Option (C) $R_f \in [-2,0]$

Not onto

Option (D) $R_f \in [1,3]$

Not onto

12. Let $f : R \to R$ be defined as $f(x) = 3^{-|x|} - 3^x + \text{sgn}(e^{-x}) + 2$
(where sgn x denotes signum function of x). Then which one of the following is correct ?

(A) f is injective but not surjective (B) f is surjective but not injective

(C) f is injective as well as surjective (D) f is neither injective nor surjective

माना $f : R \to R$ में परिभाषित फलन $f(x) = 3^{-|x|} - 3^x + \text{sgn}(e^{-x}) + 2$ है ; (जहाँ sgn x, x के सिग्नम फलन को दर्शाता है ।) तब निम्न में से कौनसा कथन सत्य है–

(A) f एकैकी है लेकिन आच्छादक नहीं है । (B) f आच्छादक है लेकिन एकैकी नहीं है ।

(C) f एकैकी और आच्छादक है (D) f न तो एकैकी है और ना ही आच्छादक है ।

Ans. D

Sol. Given :

$$f(x) = \begin{cases} 3 & x \le 0 \\ 3^{-x} - 3^x + 3 & x > 0 \end{cases}$$

For $x < 0$; f(x) is constant. So, f(x) is many-one function.

Also, Range of f(x) is $[3, \infty)$

Therefore f(x) is neither one-one nor onto.

1. The function f(x) is defined in [0, 1] then the domain of definition of the function f[*l*n (1-x²)] is given by :

(A) $x \in \{0\}$ (B) $x \in [-\sqrt{1+e}\ -1] \cup [1+\sqrt{1+e}\ ]$

(C) $x \in (-\infty, \infty)$ (D) None of these

अन्तराल [0, 1] में परिभाषित फलन f(x) है तो फलन f[*l*n (1-x²)] का प्रान्त होगी :

(A) $x \in \{0\}$ (B) $x \in [-\sqrt{1+e}\ -1] \cup [1+\sqrt{1+e}\ ]$

(C) $x \in (-\infty, \infty)$ (D) इनमें से कोई नहीं

Ans. A

Sol. Given that Domain of f(x) is [0,1]

$x \in [0,1]$ ...(i)

for definition of f(x);

$$0 \le \ell n\left(1-x^2\right) \le 1$$

$$e^0 \le 1-x^2 \le e \Rightarrow 1 \le 1-x^2 \le e$$

$$-e \le x^2 - 1 \le -1$$

$$1-e \le x^2 \le 0$$

$$\sqrt{1-e} \le x \le 0$$

$\sqrt{1-e}$ produce –ve in square root.

So, $x \le 0$ ...(ii)

Therefore from (i) and (ii)

$$x \in \{0\}$$

2. If $f(x) = px + q$ and $f(f(f(x))) = 8x + 21$, where p and q are real numbers, then p + q equals :

यदि $f(x) = px + q$ और $f(f(f(x))) = 8x + 21$, जहां p और q वास्तविक संख्याएं हैं, तब p + q =

(A) 3 (B) 5 (C) 7 (D) 11

Ans. B

Sol. $f(px + q) = p(px + q) + q$

$f(f(f(x))) = p(p(px + q) + q) + q = p(p^2x + pq + q) + q$

$= p^3x + p^2q + pq + q = 8x + 21$

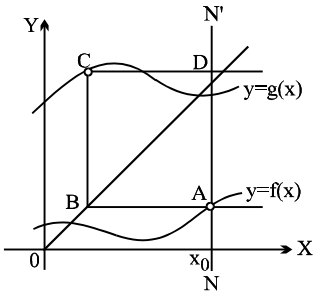
$p^3 = 8$ and $4q + 3q + q = 21$

$\Rightarrow p = 2 \qquad q = 3$

$\Rightarrow p + q = 2 + 3 = 5$

3. Given the graphs of the two functions, $y = f(x)$ and $y = g(x)$. In the adjacent figure from point A on the graph of the function $y = f(x)$ corresponding to the given value of the independent variable (say x_0), a straight line is drawn parallel to the X-axis to intersect the bisector of the first and the third quadrants at point B . From the point B a straight line parallel to the Y-axis is drawn to intersect the graph of the function $y = g(x)$ at C. Again a straight line is drawn from the point C parallel to the X-axis, to intersect the line NN′ at D . If the straight line NN′ is parallel to Y-axis, then the co-ordinates of the point D are :

$y = f(x)$ तथा $y = g(x)$ फलन का आरेख दिखाये हुये चित्रानुसार दिया गया है। बिन्दु A से आरेख $y = f(x)$ पर $(x = x_0)$ से एक रेखा x अक्ष के समान्तर खीची जाती है जो कि प्रथम चतुर्थांश तथा तृतीय चतुर्थांश अर्धक को बिन्दु B पर मिलती है। B से एक रेखा y अक्ष के समान्तर खीची जाती है जो कि आरेख $y = g(x)$ को बिन्दु C पर मिलती है पुनः बिन्दु C से एक रेखा x अक्ष के समान्तर खींची जाती है जो कि NN′ रेखा के बिन्दु D पर मिलती है। यदि रेखा NN′, Y-अक्ष के समान्तर हो तो D का निर्देशांक होगा :



(A) $f(x_0), g(f(x_0))$ (B) $x_0, g(x_0)$ (C) $x_0, g(f(x_0))$ (D) $f(x_0), f(g(x_0))$

Ans. C

Sol. According to diagram

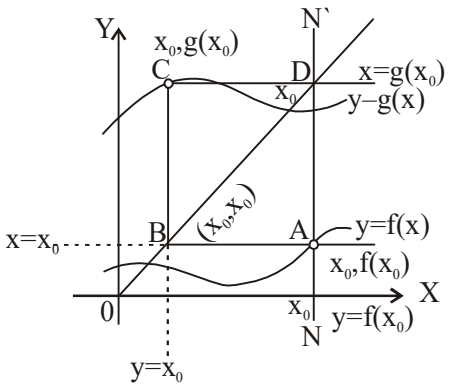
point a = $(x_0, f(x_0))$

point B = $(f(x_0), f(x_0))$

point C = $(f(x_0), f(f(x_0)))$

point D = $(x_0, g(f(x_0)))$

Therefore correct option is C.



4. Which of the following is an even function?

निम्न में से कौनसा फलन सम फलन होगा?

(A) $x\frac{a^x-1}{a^x+1}$ (B) $\tan x$ (C) $\frac{a^x-a^{-x}}{2}$ (D) $\frac{a^x+1}{a^x-1}$

Ans. A

Sol. Option (A) $f(x)=x\frac{a^x-1}{a^x+1}$

$$f(-x)=\frac{-x\left(1-a^x\right)}{\left(1+a^x\right)}=\frac{x\left(a^x-1\right)}{a^x+1}$$

Clearly f(x) = f(–x)

f(x) is even function.

5. Which of the following function is an odd function :

(A) $f(x)=\sqrt{1+x+x^2}-\sqrt{1-x+x^2}$ (B) $f(x)=x\left(\frac{a^x+1}{a^x-1}\right)$

(C) $f(x)=\log\left(\frac{1-x}{1+x^2}\right)$ (D) f(x) = k (constant)

निम्न में से कौनसा विषम फलन होगा :

(A) $f(x)=\sqrt{1+x+x^2}-\sqrt{1-x+x^2}$ (B) $f(x)=x\left(\frac{a^x+1}{a^x-1}\right)$

(C) $f(x)=\log\left(\frac{1-x}{1+x^2}\right)$ (D) f(x) = k (अचर)

Ans. A

Sol. Option (A) $f(x)=\sqrt{1+x+x^2}-\sqrt{1-x+x^2}$

$$f(-x)=\sqrt{1-x+x^2}-\sqrt{1+x+x^2}$$

$$f(-x)=-\left(\sqrt{1+x+x^2}-\sqrt{1-x+x^2}\right)=-f(x)$$

Clearly f(x) is odd function.

6. Let the function $f(x) = 3x^2 - 4x + 8\log(1 + |x|)$ be defined on the interval [0, 1]. The even extension of f(x) to the interval [–1, 0] is :

माना फलन $f(x) = 3x^2 - 4x + 8\log(1 + |x|)$ अन्तराल [0,1] में परिभाषित है तो f(x) का अन्तराल [–1, 0] में सम प्रसार होगा :

(A) $3x^2 + 4|x| + 8\log(1 + |x|)$ (B) $3x^2 - 4x + 8\log(1 + |x|)$

(C) $3x^2 + 4x - 8\log(1 + |x|)$ (D) $3x^2 - 4x - 8\log(1 + |x|)$

Ans. A

Sol. Given the function $f(x) = 3x^2 - 4x + 8\log(1 + |x|)$ be defined on the interval [0,1].

Now we know, x^2 is an even function $\forall x \in \mathbb{R}$.

But x in an odd function for $x \in [-1,1]$ to make it even we are to extend it to |x|.

And $\log(1 + |x|)$ is also an even function $\forall x \in \mathbb{R}$.

So the resulting extended even function will be $3x^2 - 4|x| + 8\log(1 + |x|)$.

7. Let f (x) be an odd function defined on (– ∞, ∞) and g (x) be an even function defined on (– ∞, ∞). If $f(x) - g(x) = x^2 + 5x + 7$, then g (2) is equal to :

माना f (x) अन्तराल (– ∞, ∞) में परिभाषित विषम फलन है तथा g (x) अन्तराल (– ∞, ∞) में परिभाषित समफलन है तथा $f(x) - g(x) = x^2 + 5x + 7$ हो तो g (2) का मान होगा :

(A) – 13 (B) 9 (C) 7 (D) – 11

Ans. D

Sol. f(x) is an odd function.

$\therefore -f(x) = f(-x)$

g(x) is an even function.

$\therefore g(-x) = g(x)$

Now, $f(x) - g(x) = x^2 + 5x + 7$ ...(1)

Put $x = -x$ in (1)

$$f(-x) - g(-x) = (-x)^2 + 5(-x) + 7$$

$$-f(x) - g(x) = x^2 - 5x + 7 \quad ...(2)$$

Adding (1) & (2) we get

$$f(x) - g(x) - f(x) - g(x) = x^2 + 5x + 7 + x^2 - 5x + 7$$

$-2g(x) = 2x^2 + 14 = 2\left(x^2 + 7\right)$

$g(x) = -\left(x^2 + 7\right)$

$g(2) = -(4 + 7) = -11$

Paragraph for question nos. 8 to 10

Let $f : [1, \infty) \to [a, \infty)$ defined by $f(x) = 2^{2x^2-4x}$ and $g : \left[\frac{\pi}{2}, \pi\right] \to [c, d]$ defined by $g(x) = \frac{\cos x + 3}{\cos x + 2}$ are two invertible functions.

यदि f और g दो फलन जिनके प्रतिलोम सम्भव है, इस प्रकार से परिभाषित किये जाते हैं $f : [1, \infty) \to [a, \infty)$; $f(x) = 2^{2x^2-4x}$ और $g: \left[\frac{\pi}{2}, \pi\right] \to [c, d]$; $g(x) = \frac{\cos x + 3}{\cos x + 2}$ तब।

8. The value of a is equal to :

'a' का मान है :

(A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) 1

Ans. B

Sol. $y = 2^{2x^2-4x}$

$\log_2 y = 2x^2 - 4x$

$= 2(x^2 - 2x + 1 - 1)$

$= 2(x - 1)^2 - 2$

$\Rightarrow 2 + \log_2 y = 2(x-1)^2$

$1 + \sqrt{\frac{2 + \log_2 y}{2}} = x$

So, $\frac{2 + \log_2 y}{2} \geq 0$

$2 + \log_2 y \geq 0$

$\log_2 y \geq -2$

$y \geq 2^{-2}$

$y \geq \frac{1}{4}$

9. Range of the function $h(x) = 4ax^2 + 2cx + d$ is equal to :

फलन $h(x) = 4ax^2 + 2cx + d$ का परिसर :

(A) $\left[\frac{-1}{4}, \infty\right)$ (B) $\left[\frac{1}{4}, \infty\right)$ (C) $\left(-\infty, \frac{-1}{4}\right]$ (D) $\left(-\infty, \frac{1}{4}\right]$

Ans. A

Sol. $\frac{y}{1} = \frac{\cos x + 3}{\cos x + 2}$

$\frac{y+1}{y-1} = \frac{\cos x + 3 + \cos x + 2}{\cos x + 3 - \cos x - 2}$

$\frac{y+1}{y-1} = 2\cos x + 5$

$\frac{y+1-5y+5}{y-1} = 2\cos x = \frac{-4y+6}{y-1}$

$\cos x = \frac{-2y+3}{y-1} \Rightarrow -1 \leq \frac{-2y+3}{y-1} \leq 0$

$\frac{-2y+3+y-1}{y-1} \geq 0$

$\frac{-y+2}{y-1} \geq 0$ and $\frac{\left(y-\frac{3}{2}\right)}{(y-1)} \geq 0$

$\frac{y-2}{y-1} \leq 0$ $y \in (-\infty, 1) \cup \left[\frac{3}{2}, \infty\right)$

$\frac{y-2}{y-1} \leq 0$

$c = \frac{3}{2}, d = 2$

$h(x) = x^2 + 3x + 2$

$= x^2 + 3x + \frac{9}{4} + 2 - \frac{9}{4}$

$= \left(x + \frac{3}{2}\right)^2 - \frac{1}{4}$

So, range of $h(x) \in \left[\frac{1}{4}, \infty\right)$

10. $f^{-1}(x)$ is equal to :

$f^{-1}(x)$ बराबर है :

(A) $\frac{4 + \sqrt{16 + 8\log_2 x}}{4}$ (B) $\frac{4 - \sqrt{16 + 8\log_2 x}}{4}$ (C) $\frac{4 + \sqrt{4 + \log_2 x}}{2}$ (D) $\frac{4 - \sqrt{4 + \log_2 x}}{2}$

Ans. A

Sol. $x = 1 + \sqrt{\frac{2 + \log_2 y}{2}}$

So, $f^{-1}(x) = 1 + \sqrt{\frac{2 + \log_2 x}{2} \times \frac{8}{8}}$

$= 1 + \frac{\sqrt{16 + 8\log_2 x}}{4}$

$f^{-1}(x) = \frac{4 + \sqrt{16 + 8\log_2 x}}{4}$

11. Let, $f(x) = x + 1, \quad x \le 1$ $\quad g(x) = x^2, \quad -1 \le x < 2$

$= 2x + 1, \quad 1 < x \le 2$ $\quad = x + 2, \quad 2 \le x \le 3$

Find fog

माना, $f(x) = x + 1, \quad x \le 1$ $\quad g(x) = x^2, \quad -1 \le x < 2$

$= 2x + 1, \quad 1 < x \le 2$ $\quad = x + 2, \quad 2 \le x \le 3$

fog ज्ञात कीजिए।

Ans. $f(g(x)) = \begin{cases} x^2+1, & -1 \le x \le 1 \\ 2x^2+1, & 1 < x \le \sqrt{2} \end{cases}$

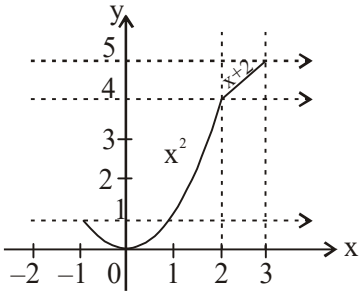
Sol. $f(g(x)) = \begin{cases} g(x)+1, & g(x) \le 1 \\ 2g(x)+1, & 1 < g(x) \le 2 \end{cases}$

Here, g(x) becomes the variable that means we should draw the graph.

It is clear that $g(x) \le 1$; $\forall\ x \in [-1, 1]$

and $1 < g(x) \le 2$; $\forall\ x \in (1, \sqrt{2}]$

$\Rightarrow$ $f(g(x)) = \begin{cases} x^2+1, & -1 \le x \le 1 \\ 2x^2+1, & 1 < x \le \sqrt{2} \end{cases}$



12. Match the following columns :

Column-I	**Column-II**
(A) Let $f: R \to R$ defined as $f(x) = e^{\text{sgn}\,x} + e^{x^2}$, where sgn x denotes signum function of x, then f(x) is	(P) Odd
(B) Let $f: (-1, 1) \to R$ defined as $f(x) = x[x^4] + \frac{1}{\sqrt{1-x^2}}$ where [x] denotes greatest integer less than or equal to x, then f(x) is	(Q) Even
(C) Let $f: R \to R$ defined as $f(x) = \frac{x(x+1)(x^4+1)+2x^4+x^2+2}{x^2+x+1}$ then f(x) is	(R) Neither odd nor even
(D) Let $f: R \to R$ defined as $f(x) = x + 3x^3 + 5x^5 + ... + 101x^{101}$ then f(x) is	(S) One-One
	(T) Many-One

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) माना $f: R \to R$ इस प्रकार है कि $f(x) = e^{\text{sgn}\,x} + e^{x^2}$, जहाँ sgn x सिग्नम फलन दर्शाता है तो f(x) होगा	(P) विषम
(B) माना $f: (-1, 1) \to R$ इस प्रकार है कि $f(x) = x[x^4] + \frac{1}{\sqrt{1-x^2}}$ जहाँ [x] महत्तम पूर्णांक फलन दर्शाता है तो f(x) होगा	(Q) सम

(C) माना $f : R \to R$ इस प्रकार है कि $f(x) = \dfrac{x(x+1)(x^4+1)+2x^4+x^2+2}{x^2+x+1}$ तो $f(x)$ होगा। (R) न तो सम न ही विषम

(D) माना $f : R \to R$ इस प्रकार है कि $f(x) = x + 3x^3 + 5x^5 + ... + 101x^{101}$ तो $f(x)$ होगा। (S) एकैकी

(T) बहुएकैकी

Ans. (A) R, T ; (B) Q, T ; (C) R, T ; (D) P, S

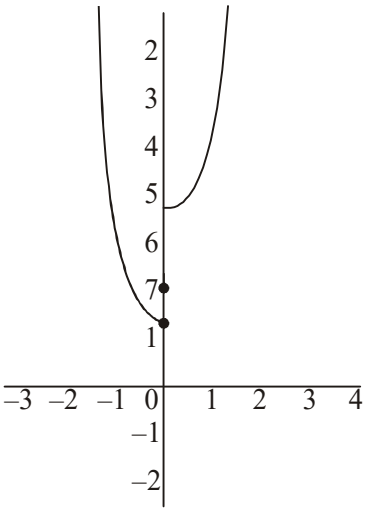
Sol. (A) $f(x) = e^{\text{sgn}(x)} + e^{x^2}$

$f(-x) = e^{\text{sgn}(-x)} + e^{-x^2}$

$f(x) - f(-x) \neq 0$

$f(x) + f(-x) \neq 0$

Therefore function is neither odd nor even.



$$f(x) = e^{\frac{|x|}{x}} + e^{x^2} = \begin{cases} \dfrac{1}{e} + e^{x^2} & x < 0 \\ 2 & x = 0 \\ e + e^{x^2} & x > 0 \end{cases}$$

Clear from definition and graph that function is many one.

(B) Since, $-1 < x < 1$

$0 < x^2 < 1$

$0 < x^4 < 1$. So, $[x^4] = 0$

$$f(x) = x \times 0 + \frac{1}{\sqrt{1-x^2}} = \frac{1}{\sqrt{1-x^2}}$$

$$f(-x) = -x \times 0 + \frac{1}{\sqrt{1-x^2}} = \frac{1}{\sqrt{1-x^2}}$$

$f(x) = f(-x)$. Hence, $f(x)$ is even function.

Also, f(x) is many-one function (as at $x = \frac{1}{2}$ & $-\frac{1}{2}$ value of f(x) are same)

(C) $f(x) = \frac{\left(x^2 + x\right)\left(x^4 + 1\right) + 2x^4 + x^2 + 2}{\left(x^2 + x + 1\right)}$

$$f(-x) = \frac{-\left(x - x^2\right)\left(x^4 + 1\right) + 2x^4 + x^2 + 2}{\left(x^2 - x + 1\right)}$$

Neither even nor odd

Also its an even degree function. Hence many one.

(D) Given : $f(x) = x + 3x^3 + 5x^5 + ... + 101x^{101}$

$f(-x) = -[x + 3x^3 + 5x^5 + ... + 101x^{101}] = -f(x)$

$\therefore$ f(x) is odd function.

As its odd degree function. Hence one-one.

1. If $f(x) = x^3 - 1$ and domain of $f = \{0,1,2,3\}$, then domain of f^{-1} is :

यदि $f(x) = x^3 - 1$ तथा f का प्रान्त $= \{0,1,2,3\}$ तो f^{-1} का प्रान्त होगा :

(A) $\{0,1,2,3\}$ (B) $\{1,0,-7,-26\}$ (C) $\{-1,0,7,26\}$ (D) $\{0,-1,-2,-3\}$

Ans. C

Sol. Given : domain of $f = \{0,1,2,3\}$

Range of f = Domain of f^{-1}.

$\{-1,0,7,26\}$ = Domain of f^{-1}.

2. The inverse of the function $y = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$ is :

फलन $y = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$ का प्रतिलोम होगा :

(A) $\dfrac{1}{2}\log\dfrac{1+x}{1-x}$ (B) $\dfrac{1}{2}\log\dfrac{2+x}{2-x}$ (C) $\dfrac{1}{2}\log\dfrac{1-x}{1+x}$ (D) $2\log(1+x)$

Ans. A

Sol. $\dfrac{y}{1} = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$

using componendo dividendo

$$\frac{y+1}{y-1} = \frac{e^x - e^{-x} + e^x + e^{-x}}{e^x - e^{-x} - e^x - e^{-x}}$$

$$\frac{y+1}{y-1} = \frac{2\cdot e^x}{-2\cdot e^{-x}} = -\frac{e^x}{e^{-x}}$$

$$\frac{y+1}{y-1} = -e^{2x} \Rightarrow e^{2x} = -\frac{y+1}{y-1}$$

$e^{2x} = \dfrac{y+1}{1-y}$ taking log both sides

$$2x = \ell n\left(\frac{1+y}{1-y}\right) \Rightarrow x = \frac{1}{2}\ell n\left(\frac{1+y}{1-y}\right)$$

3. Let $f : N \to N$ where $f(x) = x + (-1)^{x-1}$ then f is :

(A) Inverse of itself (B) even function (C) periodic (D) identity

माना $f : N \to N$ जहाँ $f(x) = x + (-1)^{x-1}$ तो f होगा :

(A) स्वयं का प्रतिलोम (B) समफलन (C) आवर्ती फलन (D) तत्समक फलन

Ans. A

Sol. $f(x) = \begin{cases} x-1, & x \text{ is even} \\ x+1, & x \text{ is odd} \end{cases}$, which is clearly are one-one and onto.

Hence given function is inverse of itself.

$f(x) = \begin{cases} x-1, & x \text{ सम है} \\ x+1, & x \text{ विषम है} \end{cases}$, जोकि स्पष्टतया एकैकी और आच्छादक है।

अतः दिया गया फलन स्वयं का प्रतिलोम है।

4. Which of the following functions is inverse of itself :

(A) $f(x) = \frac{1-x}{1+x}$ (B) $g(x) = 5^{\log x}$ (C) $h(x) = 2^{x(x-1)}$ (D) None of these

निम्न में से कौनसा फलन स्वयं का प्रतिलोम है :

(A) $f(x) = \frac{1-x}{1+x}$ (B) $g(x) = 5^{\log x}$ (C) $h(x) = 2^{x(x-1)}$ (D) इनमें से कोई नहीं

Ans. A

Sol. Check all options :

(A) $y = \frac{1-x}{1+x}$

replace x & y

$x = \frac{1-y}{1+y}$

$\Rightarrow f^{-1}(x) = \frac{1-x}{1+x}$ f(x) is inverse of itself.

(B) $y = 5^{\ell nx}$

replace x & y $\Rightarrow x = 5^{\ell ny}$

$\Rightarrow \log_5 x = \ell ny$

$\Rightarrow y = e^{\log_5 x} \neq f(x)$

(C) $y = 2^{x(x-1)}$

replace x & y mutually.

$x = 2^{y(y-1)}$

$\log_2 x = y(y-1) \neq f(x)$

5. If $f(x) = 2x^3 + 7x - 5$ then $f^{-1}(4)$ is :

(A) equal to 1 (B) equal to 2 (C) equal to 1/3 (D) non existent

यदि $f(x) = 2x^3 + 7x - 5$ हो, तो $f^{-1}(4)$:

(A) 1 के बराबर होगा (B) 2 के बराबर होगा (C) 1/3 के बराबर होगा (D) अस्तित्व विहीन

Ans. A

Sol. Range of f = Domain of f^{-1}

So, $4 = 2x^3 + 7x - 5$

$2x^3 + 7x - 9 = 0$

$x = 1$ is solution of above equation.

6. If $f : R \to R$ is a function satisfying the property $f(x+1) + f(x+3) = 2 \ \forall \ x \in R$ then the period (may not be fundamental period) of f(x) is :

यदि फलन $f : R \to R$ गुणधर्म $f(x+1) + f(x+3) = 2 \ \forall \ x \in R$ को सन्तुष्ट करता है तो फलन f का आवर्तकाल होगा :

(A) 3 (B) 4 (C) 7 (D) 6

Ans. B

Sol. $f : R \to R$

$\therefore f(x+1) + f(x+3) = 2 \qquad \forall x \in R$

for $x = 0$ $f(1) + f(3) = 2$ ...(1)

for $x = 2$ $f(3) + f(5) = 2$ ...(2)

By solving (i) & (ii)

$f(1) = f(5)$

$\because f(1) = f(5)$

$f(1) = f(1+4)$

$\therefore$ period of function = 4.

7. The fundamental period of the function
$f(x) = x + a - [x + b] + \sin \pi x + \cos 2\pi x + \sin 3\pi x + \cos 4\pi x + \ldots + \sin (2n-1)\pi x + \cos 2\pi n\, x$ for every $a, b \in R$ is : (where [.] denotes the greatest integer function)

निम्न फलन का आवर्तकाल होगा
$f(x) = x + a - [x + b] + \sin \pi x + \cos 2\pi x + \sin 3\pi x + \cos 4\pi x + \ldots + \sin (2n-1)\pi x + \cos 2\pi n\, x$ सभी $a, b \in R$ के लिये : (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

(A) 2 (B) 4 (C) 1 (D) 0

Ans. A

Sol. $f(x) = x + a - (x + b - \{x + b\}) + (\sin\pi x + \sin 3\pi x + \ldots + \sin(2n-1)\pi x) + (\cos 2\pi x + \cos 4\pi x + \ldots + \cos 2\pi n x)$

$$\text{Period of fx)} = \underbrace{(a-b)-\{x+b\}}_{\text{period}=1} + \text{LCM}\left(2, \frac{2}{3}, \frac{2}{5}, \ldots\ldots, \frac{2}{(2n-1)}\right) + \text{LCM}\left(1, \frac{1}{2}, \frac{1}{6}, \ldots\ldots, \frac{1}{n}\right)$$

$$\text{Period of f(x)} = \text{LCM of } 1, \left(\text{LCM of } 2, \frac{2}{3}, \frac{2}{5}, \ldots, \frac{2}{2n-1}\right), \left(\text{LCM of } 1, \frac{1}{2}, \frac{1}{6}, \ldots, \frac{1}{n}\right)$$

Period of f(x) = 2.

8. Let $f(x) = \sin \sqrt{[a]}\, x$ (where [ ] denotes the greatest integers function). If f is periodic with fundamental period π, then a belongs to :

माना $f(x) = \sin \sqrt{[a]}\, x$ (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है). यदि f का मूलभूत आवर्तकाल π, है तो a किस अन्तराल में होगा :

(A) [2, 3) (B) {4, 5} (C) [4, 5] (D) [4, 5)

Ans. D

Sol. $f(x) = \sin \sqrt{[a]}\ x$

period of $\sin x = 2\pi$

$$\Rightarrow \text{ period of } f(x) = \frac{2\pi}{\sqrt{[a]}} = \pi$$

$\Rightarrow \sqrt{[a]} = 2 \Rightarrow [a] = 4 \Rightarrow a \in [4, 5)$

9. Period of $f(x) = e^{\cos\{x\}} + \sin \pi [x]$ is : (where, [.] and { } denote the greatest integer function and fractional part of function respectively)

$f(x) = e^{\cos\{x\}} + \sin \pi [x]$ का आवर्तकाल होगा (जहाँ [.] तथा { } क्रमशः महत्तम पूर्णांक फलन तथा भिन्नातम भाग दर्शाते है)

(A) 1 (B) 2 (C) π (D) 2

Ans. A

Sol. $f(x) = e^{\cos\{x\}} + \sin \pi [x]$

period of $e^{\cos\{x\}} = 1$

period of $\sin \pi[x]$ = no fundamental period

$\therefore$ period of $f(x)$ = L.C.M. of $\{1, \text{N.D.}\}$

= 1

10. If $f(x) = \cos(ax) + \sin(bx)$ is periodic, then which of the followings is false :

(A) a and b both are rational

(B) non-periodic if a is rational but b is irrational

(C) non-periodic if a is irrational but b is rational

(D) none of these

यदि $f(x) = \cos(ax) + \sin(bx)$ आवर्ती फलन है तो निम्न में से कौनसा असत्य होगा :

(A) a तथा b दोनों परिमेय है

(B) यदि a परिमेय है तथा b अपरिमेय है तो फलन आवर्ती नहीं होगा

(C) यदि b परिमेय है तथा a अपरिमेय है तो फलन आवर्ती नहीं होगा

(D) कोई नहीं

Ans. D

Sol. $f(x) = \sin bx + \cos ax$

for periodic,

L.C.M. of $\left(\frac{2\pi}{b}, \frac{2\pi}{a}\right)$

$$= \frac{\text{L.C.M. of } (2\pi, 2\pi)}{\text{H.C.F. of } (a, b)}$$

here a, b must be integer

$$= \frac{2\pi}{1} = 2\pi$$

11. The function $f(x) = 2(x - [x]) + \sin^2\pi (x - [x])$ is : (Where [.] denotes greatest integer function)

(A) Non periodic (B) periodic with period 1

(C) periodic with period 2 (D) None

फलन $f(x) = 2(x - [x]) + \sin^2\pi (x - [x])$ है : (जहाँ [.] महत्तम पूर्णांक फलन दर्शाता है)

(A) अनावर्ती (B) आवर्ती जहाँ आवर्तकाल 1 हो

(C) आवर्ती जहाँ आवर्तकाल 2 हो (D) कोई नहीं

Ans. B

Sol. $f(x) = 2\{x\} + \sin^2 \pi\{x\}$

Period of $2\{x\}$ is 1

Period of $\sin^2 \pi\{x\}$ is $\frac{\pi}{\pi} = 1$

Period of f(x) is 1.

12. Column I contains functions and column II contains their properties. One or more entries of Column-II can be matches with one entry of column-I :

Column-I	**Column-II**
(A) $f(x) = \sin^2 2x - 2\cos^2 x$	(P) Many one but not even function
(B) $f(x) = \sqrt{\log_{0.5} \sec(\sin^2 x)}$	(Q) Both many one and even function
(C) $f(x) = \text{sgn}\left(\frac{1-x^2}{1+x^2}\right) + \cos 2\pi \left[x^2 - 5x + 6\right]$ where sgn x and [x] denote signum function of x and largest integer less than or equal to x	(R) Periodic but not odd function
(D) $f(x) = \frac{\cos x}{\lvert\cos x\rvert} + \cos\left(\frac{x^2 - x}{x - 1}\right)$	(S) Range contains atleast one integer but not more than three integers.
	(T) Bounded function

स्तम्भ I में फलन है एवं स्तम्भ II में उनके गुणधर्म है। स्तम्भ II के एक या एक से अधिक अवयवों को स्तम्भ I के किसी एक अवयव से मिलान कीजिये। :

Column-I	**Column-II**
(A) $f(x) = \sin^2 2x - 2\cos^2 x$	(P) बहुएकैकी लेकिन सम फलन नहीं
(B) $f(x) = \sqrt{\log_{0.5} \sec(\sin^2 x)}$	(Q) बहुएकैकी तथा सम फलन दोनों
(C) $f(x) = \text{sgn}\left(\frac{1-x^2}{1+x^2}\right) + \cos 2\pi\left[x^2 - 5x + 6\right]$ (जहाँ [x], महत्तम पूर्णांक फलन को निरुपित करता है एवं sgn x सिग्नम फलन को दर्शाता है)	(R) आवर्ती लेकिन विषम फलन नहीं
(D) $f(x) = \frac{\cos x}{\lvert\cos x\rvert} + \cos\left(\frac{x^2 - x}{x-1}\right)$	(S) परिसर में कम से कम एक पूर्णांक लेकिन तीन से अधिक पूर्णांक नही
	(T) परिबद्ध फलन

Ans. (A) Q, R, S, T ; (B) Q, S, T; (C) Q, S, T ; (D) P, S, T

Sol.

13. Find the inverse of the following function : $f(x) = \begin{cases} x, & x < 1 \\ x^2, & 1 \le x \le 4 \\ 8\sqrt{x}, & x > 4 \end{cases}$

निम्न फलन का प्रतिलोम ज्ञात कीजिये : $f(x) = \begin{cases} x, & x < 1 \\ x^2, & 1 \le x \le 4 \\ 8\sqrt{x}, & x > 4 \end{cases}$

Ans.

Sol. Case I स्थिति–I $y = x$ $x < 1$

$x = y$ $y < 1$

$f^{-1}(x) = x$ $x < 1$

Case II स्थिति–II $y = x^2$ $1 \le x \le 4$

$x^2 = y$ $1 \le y \le 16$

$x = \sqrt{y}$ $1 \le y \le 16$

$f^{-1}(x) = \sqrt{x}$ $1 \le x \le 16$

Case III स्थिति–III $y = 8\sqrt{x}$ $x > 4$

$$x = \frac{y^2}{64}\ y > 16$$

$$f^{-1}(x) = \frac{x^2}{64} \qquad x > 16$$

14. Find the period of the following function $f(x) = e^{x-[x]} + |\cos \pi x| + |\cos 2\pi x| + ... + |\cos n\pi x|$, [.] is greatest integer function.

निम्न फलन का आवर्तकाल ज्ञात कीजिये। $f(x) = e^{x-[x]} + |\cos \pi x| + |\cos 2\pi x| + ... + |\cos n\pi x|$, [.] महत्तम पूर्णांक फलन दर्शाता है।

Ans. 1

Sol. $f(x) = e^{\{x\}} + |\cos \pi x| + |\cos 2\pi x| + ... + |\cos n\pi x|$

Period of $e^{\{x\}} = 1$

Period of $|\cos \pi x| = 1$

$$|\cos 2\pi x| = \frac{1}{2}$$

$$\vdots$$

$$\vdots$$

$$|\cos n\pi x| = \frac{1}{n}$$

$$\text{LCM} = \left\{1, \frac{1}{2}, \frac{1}{3}, ..., \frac{1}{n}\right\}$$

$$= \frac{\text{LCM of } \{1,1,.....\}}{\text{HCF of } \{1,2,.....,n\}} = \frac{1}{1} = 1$$